

MATH 1564 — Notes 08/25

Definition: A linear equation is

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, a_2, \dots, a_n$ are coefficients, $a_i \in \mathbb{R}$; b is the right hand side, $b \in \mathbb{R}$; and $x_1, x_2, \dots, x_n$ are variables (unknowns), $x_i \in \mathbb{R}$.

Linear algebra can be done with finite fields, complex numbers. Real number solutions can be better visualized.

Example: $2x_1 + 3x_2 = 6$ is a line in $\mathbb{R}^2$.

In general, algebra tends to work even when geometry is hard to understand.

Example: $2x_1 + 3x_2 + 0x_3 = b$ is a plane in $\mathbb{R}^3$ with normal vector

$$\begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}.$$

We are interested in systems of linear equations. For example,

$$\begin{aligned} 2x_1 + 3x_2 + 0x_3 &= 6 \\ -x_1 + 2x_2 - x_3 &= 2 \end{aligned}$$

Definition: The system

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{cases}$$

is a system of m linear equations with unknowns $x_1, \dots, x_n$.

Questions:

1. How do we describe the solutions of a system?
2. How many solutions can a system have, and how do we decide this?
3. Why do we care about linear systems at all? (Taylor expansion: a complex system can be reduced to a linear approximation.)

A linear equation can have no solutions.

Heuristic on $\mathbb{R}^2$: For two linear equations, the corresponding lines either intersect, are parallel, or are collinear, so respectively there is 1 solution, no solution, or infinitely many solutions.

Goal: Develop an algorithm for solving a linear system that can be made into a computer program.

Note that only the numbers a_{ij} , b_i matter, and the indices i, j encode their position, so we write the augmented matrix

$$[A \mid b] = \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ & \ddots & & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array} \right].$$

The matrix encodes the system. A is the matrix of coefficients, and b is the right hand side of the system.

Strategy: Do row operations on $[A \mid b]$ to bring it to a simpler form.

We write the system as $Ax = b$ where $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$.

Elementary row operations:

1. **Replacement:** add a multiple of a row to another row.
2. **Interchange:** swap two rows.
3. **Scaling:** multiply a row by a nonzero number.

Key feature: Each row operation is reversible, so no information is lost.

Theorem 1: If $[R \mid d]$ is obtained from $[A \mid b]$ by a sequence of row operations, then $Ax = b \iff Rx = d$.

Proof: Argue by induction on $r =$ the number of operations.

Base case: $r = 0$.

Inductive step $r \rightarrow r + 1$: both sides apply the operation. Cases:

1. Replacement
2. Interchange
3. Scaling

How to simplify $[A \mid b]$ by row reduction: Aim toward RREF = row reduced echelon form.

Example: A matrix in RREF looks like (where $*$ = any number):

$$\begin{bmatrix} 0 & 1 & * & 0 & * & 0 & 0 & * \\ 0 & 0 & 0 & 1 & * & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Definition: A matrix is in RREF if:

1. All zero rows (if any) are at the bottom.
2. The leftmost nonzero entry in a row equals 1 (called a pivot) and is to the right of the pivots in the rows above.
3. Pivots are the only nonzero entries in their columns.

Examples: Is each matrix in RREF?

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \text{ No.} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ No.} \quad \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ Yes.}$$

Pivotal position, pivot.

Row reduction algorithm:

1. Cover zero columns on the left.
2. Look at the leftmost nonzero column. Pick a nonzero entry in the column.
 - (a) Move it to the top row.
 - (b) Scale it to be 1.
 - (c) Use row replacement to make all other entries in the column = 0.
3. Cover the column and the top row to get a smaller matrix, and repeat the process.

Example:

$$\begin{bmatrix} 2 & 4 & 6 \\ -1 & -2 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ (RREF)}$$

Alternatively, using different row operations on the same starting matrix:

$$\begin{bmatrix} 2 & 4 & 6 \\ -1 & -2 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ -1 & -2 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ (RREF)}$$

Example: Given

$$[R \mid d] = \left[\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

this encodes

$$\begin{aligned} x_1 + 2x_2 + 0x_3 &= 3 \\ 0x_1 + 0x_2 + 1x_3 &= 2 \end{aligned}$$

so $x_3 = 2$ and $x_1 + 2x_2 = 3$, i.e., $x_1 = 3 - 2x_2$.

x_1 and x_3 are **pivotal variables**; x_2 is **free** (the opposite of a pivotal variable).

Solution:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 - 2x_2 \\ x_2 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

This is a line in $\mathbb{R}^3$ passing through $\begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$ in the direction of $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$.